Assignment 2 Report: Neural Architectures for Misogyny Detection

Karim Abu-Shams, Ainhoa Del Rey and Iñigo Goikoetxea

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1 Introduction

This report analyzes the performance of neural sequence models for misogyny detection on the MAMI dataset (SemEval-2022 Task 5). We compare a **BiLSTM** model (trained from scratch with FastText embeddings) against a **BERT** model (fine-tuned from `bert-base-uncased`). The goal is to evaluate their effectiveness, efficiency, and robustness, specifically addressing the challenges of overfitting in small datasets and the impact of text preprocessing on model performance.

2 Methodology & Models

2.1 BiLSTM Architecture

- **Embeddings**: Pre-trained FastText (300d), frozen during training to prevent overfitting.
- **Structure**: Bidirectional LSTM (2 layers, 256 hidden units).
- **Regularization**:
  - `LockedDropout` (0.5) applied between layers (variational dropout).
  - `WordDropout` (0.2) dropping entire word embeddings.
  - Weight Decay (10^{-4}) and Label Smoothing (0.1).

2.2 BERT Architecture

- **Base Model**: `bert-base-uncased` (110M params).
- **Fine-tuning Strategy**: **Gradual Unfreezing**.
  - Epoch 0: Train only classifier head.
  - Epoch 1: Unfreeze last 2 layers.
  - Epoch 2: Unfreeze last 4 layers.
  - Epoch 3+: Unfreeze all layers.
- **Regularization**: Dropout (0.4) in the classification head, Label Smoothing (0.1).

2.3 Addressing Overfitting: Initial Challenge

A major challenge in this task was the small dataset size (10,000 training samples) relative to the capacity of the models, particularly BERT (110M parameters). Our **initial experiments without regularization** showed severe overfitting:

- Training F1 rapidly reached ∼0.99 while validation F1 plateaued at ∼0.60.
- The gap between training and validation performance indicated the models were memorizing the training data rather than learning generalizable patterns.

To mitigate this, we progressively increased regularization:

- **Aggressive Dropout**: Increased from the standard 0.1 to **0.5** for LSTM and **0.4** for BERT. This was necessary to prevent the models from overfitting on the small dataset.
- **Variational Dropout**: For LSTM, applying the same dropout mask across time steps (`LockedDropout`) prevented the model from memorizing specific patterns.
- **Gradual Unfreezing**: For BERT, progressively unfreezing layers prevented catastrophic forgetting and allowed the model to adapt to the task without destroying pre-trained knowledge.
- **Label Smoothing (0.1)**: Softened the target distribution to discourage overconfident predictions.

After applying these techniques, the training-validation gap was successfully reduced (Training F1 ∼0.74, Validation F1 ∼0.75), but the test performance remained below our expectations.

3 Experimental Results

3.1 Architecture Comparison (Experiment 1)

We compare the two neural models against the **winner of Assignment 1**: Logistic Regression with FastText embeddings (300d, averaged), which achieved 0.6395 F1-Macro on the official test set.

Model	Representation	Test F1	Val F1	Params	Time
A1 Winner: LR + FastText	Dense (300d)	**0.6395**	0.7252 (CV)	301	< 1s
A1 Runner-up: SVM + TF-IDF	Sparse (bigrams)	0.6360	0.7757 (CV)	5,001	< 1s
BiLSTM	FastText (300d)	0.6068	0.7476	2.7M	58s
BERT (Gradual Unfreeze)	Contextual	0.6294	0.8289	110M	145s

Table 1: Initial performance comparison—BERT underperforms the A1 baselines.

Problem Identified: Despite being a more powerful model with 110M parameters and achieving excellent validation F1 (∼0.83), BERT **failed to beat the simple Logistic Regression baseline** on the official test set (0.6294 vs 0.6395). This counterintuitive result prompted a detailed investigation.

3.2 Learning Curves (Experiment 2)

We evaluated performance on 25%, 50%, 75%, and 100% of training data.

- **BERT** showed superior data efficiency, achieving **0.6173 F1** with just 25% of data, whereas LSTM struggled at **0.5792**.
- BERT's performance scales linearly with data, suggesting it would benefit significantly from more labeled examples.

3.3 Ablation Studies (Experiment 3)

- **LSTM**: The **Bi-LSTM (1 layer, 128 hidden)** performed best (0.6247 F1), suggesting simpler models generalize better on this small dataset than deeper ones (2 layers yielded 0.6068).
- **BERT**:
  - **Full Fine-tuning** achieved the highest F1 (0.6408), surpassing the baseline.
  - **Gradual Unfreezing** (0.6155) provided stability and high validation scores (0.82) but slightly lower test generalization than aggressive full fine-tuning in this specific run.
  - **DistilBERT** (0.6397) offered a competitive alternative with 40% fewer parameters, matching the baseline performance.

3.4 Error Analysis (Experiment 4)

- **Complementarity**: Neural models corrected ∼110 errors made by the baseline.
- **New Errors**: However, they also introduced ∼100 new errors, often on samples requiring deep reasoning or world knowledge (memes).
- **Confusion Matrix**: Both models show a balanced capability in detecting misogynous content, though false negatives remain the primary challenge.

3.5 Computational Cost (Experiment 5)

- **Inference Speed**: BiLSTM is ∼**8.5x faster** than BERT (9460 samples/sec vs 1117 samples/sec).
- **Memory**: BERT requires ∼3GB GPU memory, while BiLSTM needs only ∼178MB.
- **Conclusion**: BiLSTM is preferable for resource-constrained environments, while BERT is better for maximizing performance when resources allow.

4 Performance Improvement Investigation (Experiment 6)

4.1 Motivation: Why Did BERT Underperform?

The results from Experiments 1–3 revealed an unexpected paradox: **BERT (110M parameters) was unable to consistently surpass a simple Logistic Regression** baseline on the official test set. According to the MAMI SemEval-2022 paper, BERT-based systems typically achieve F1 scores in the range of 0.65–0.72 (text-only), with the competition median at 0.679 and top systems reaching 0.834 (using multimodal ensembles). Our BERT score of 0.6155–0.6294 placed us **below the 25th percentile** (Q1 = 0.649) of participating systems.

We conducted a systematic investigation to identify the root causes and potential fixes.

4.2 Root Cause Analysis

After careful inspection of the code and data pipeline, we identified the following issues:

1. **Raw Text Input (Primary Cause)**: The BERT model was receiving **unprocessed raw text**, including URLs (e.g., `http://...`), social media mentions (`@user`), meme watermarks (e.g., `memefulcom`, `quickmemecom`), and excessive punctuation. While BERT's tokenizer is designed to handle diverse text, these noisy artifacts consumed valuable tokens in the 128-token window and introduced misleading signals. In contrast, the A1 baseline used **cleaned text** where these artifacts were removed prior to feature extraction.
2. **Over-regularization (Secondary)**: While high dropout (0.4) and gradual unfreezing successfully addressed overfitting on the training set, they also potentially limited the model's capacity to learn discriminative features from the noisy raw input. The combination of label smoothing, high dropout, and gradual unfreezing may have been too conservative for the amount of useful signal in the data.
3. **Validation–Test Distribution Shift**: All configurations showed a large gap between validation F1 (∼0.82) and test F1 (∼0.63). This is a **known property of the MAMI dataset**, confirmed by the task organizers: the test set contains examples that are systematically harder and drawn from a slightly different distribution than the training set.

4.3 Systematic Comparison (Experiment 6)

To isolate the impact of each factor, we ran a controlled experiment with 5 configurations:

#	Configuration	Test F1	Val F1	Time (s)
0	A1 Winner (LR + TF-IDF Bigrams)	0.6347	N/A	< 1s
1	BERT Original (Raw + Gradual + Drop 0.4)	0.6155	0.8210	143s
2	**BERT + Clean Text (Clean + Gradual + Drop 0.4)**	**0.6654**	0.8090	141s
3	BERT + Clean + FullFT (Drop 0.1)	0.6383	0.8200	107s
4	BERT + Clean + FullFT + No Label Smoothing	0.6415	0.8180	107s

Table 2: Experiment 6: Impact of text cleaning and hyperparameter tuning on BERT.

4.4 Analysis of Improvements

Key Findings:

1. **Clean text = +5% F1** (0.6155 → 0.6654): Simply applying the same text cleaning used in Assignment 1 (removing URLs, mentions, punctuation) improved BERT by **5 percentage points**. This is the single most impactful change, confirming that preprocessing is critical even for models with sophisticated tokenizers.

2. **BERT + Clean Text now beats the baseline by +3%**: Configuration 2 achieved **0.6654 F1**, surpassing the A1 baseline (0.6347) by a meaningful margin. This is consistent with the competition results where BERT-based text-only systems achieve 0.65–0.72.

3. **Lower dropout did NOT help on the test set**: Configurations 3 and 4 (dropout 0.1, full fine-tuning) achieved higher validation F1 (∼0.82) but *lower* test F1 (0.64) than the high-dropout version (0.66). This demonstrates that the aggressive regularization is **beneficial given the train–test distribution shift**: it prevents the model from overfitting to patterns in the training distribution that do not transfer to the test set.

4. **Label smoothing has minimal impact**: Removing label smoothing (config 4 vs 3) changed test F1 from 0.6383 to 0.6415, a negligible difference within the variance of the results.

Why did clean text help so much? Meme text contains many domain-specific artifacts that are noise for the classification task:

- **Watermarks**: Strings like "memefulcom", "quickmemecom" appear in many memes regardless of misogynistic content, consuming token positions.

- **URLs and mentions**: These are irrelevant features that BERT might attend to, diluting its attention on the actual content.

- **Token budget**: With a 128-token limit, every token wasted on noise reduces the signal-to-noise ratio of the input representation.

Why didn't lower regularization help? The MAMI dataset has a significant distribution shift between training and test sets. Models with lower regularization fit the training distribution more tightly, including patterns that only appear in training (higher validation F1). The test set requires more robust, generalizable features—exactly what higher regularization encourages.

5 Conclusion

We successfully implemented, regularized, and improved neural models for misogyny detection. Our work progressed through three phases:

1. **Phase 1 – Initial Implementation**: Without regularization, BERT severely overfitted (Train F1 ∼0.99, Val F1 ∼0.60). BiLSTM showed similar patterns.

2. **Phase 2 – Regularization**: We applied aggressive dropout (0.4–0.5), gradual unfreezing, variational dropout, and label smoothing. This stabilized training but BERT still underperformed the baseline (0.6155 vs 0.6395).

3. **Phase 3 – Investigation & Fix** (Experiment 6): We identified that feeding BERT **raw text with noise** was the root cause. Applying text cleaning improved BERT from 0.6155 to **0.6654 F1**, finally surpassing the baseline by +3%.

Key Lessons Learned:

- **Preprocessing matters even for BERT**: Despite having a sophisticated tokenizer, BERT benefits from removing noise (URLs, watermarks, mentions) in domain-specific applications with small datasets.
- **Regularization is essential with distribution shift**: High dropout (0.4) + gradual unfreezing outperformed standard fine-tuning (drop 0.1) on the test set, due to the known train–test distribution shift in the MAMI dataset.
- **Validation scores can be misleading**: All BERT configurations achieved Val F1 ∼0.82 regardless of preprocessing, but test F1 varied from 0.61 to 0.67. The validation set shares the training distribution and does not reflect real-world generalization.
- **Simple baselines are strong**: The LR + TF-IDF bigrams baseline remains competitive (0.6347), demonstrating that for small datasets with explicit discriminative keywords, classical approaches are surprisingly effective.